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Batch - 4

{ LSTM Practice Question }

① Input $x_t = 0.5$
Prev hidden state $h_{t-1} = 0.2$
Prev cell state $C_{t-1} = 0.4$

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$\tanh(z) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$$

(i) forget gate: $f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$

Substitute Values:

$$f_t = \sigma(0.7(0.5) + 0.3(0.2) + 0.1)$$
$$= \sigma(0.510)$$

Sigmoid : $\frac{1}{1+e^{-0.5100}}$

$$\approx 0.6248$$

$$\{ f_t = 0.6248 \}$$

The forget gate retains approximately 62.48-1. of the prev. cell state information.

(ii) input gate: $i_t = \sigma(w_i x_t + U_i h_{t-1} + b_i)$

$i_t = \sigma(0.6(0.5) + 0.4(0.2) + 0.0)$

$i_t = \sigma(0.380)$

Sigmoid: $\frac{1}{1 + e^{-0.38}}$

$\{ i_t = 0.5938 \}$

The input gate controls how much new candidate information enters the cell state.

(iii) Candidate State: $\tilde{C}_t = \tanh(w_t x_t + U_c h_{t-1} + b_c)$

Substitute values

$\tilde{C}_t = \tanh(0.9(0.5) + 0.2(0.2) + 0.0)$

$\tilde{C}_t = \tanh(0.4900)$

$\tilde{C}_t = \frac{e^{0.49} - e^{-0.49}}{e^{0.49} + e^{-0.49}}$

$\{ \tilde{C}_t = 0.45 \}$

(iv) New cell State C_t :

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t$$

$$C_t = (0.6248)(0.4) + (0.5938)(0.4542)$$

$$\{C_t = 0.5196\}$$

The new cell state contains both retained information from the prev. state & new candidate info.

(v) output gate : O_t

$$O_t = \sigma(w_o x_t + u_o h_{t-1} + b_o)$$

$$O_t = \sigma(0.4)$$

$$O_t = \frac{1}{1 + e^{-0.4}}$$

$$\{O_t = 0.6100\}$$

(vi) Hidden State h_t

$$h_t = O_t \tanh(C_t)$$

$$h_t = 0.61 \tanh(0.5196)$$

$$\{h_t = 0.2912\}$$

(2) input Dim. D = 10
Hidden units H = 32

(a) One bias vector per gate

For each gate, the trainable parameters are:

(1) Input weight : D×H
(2) Recurrent weight : H×H
(3) Bias : H

for 1 gate : $P_{gate} = DH + H^2 + H$

$P_{gate} = (10)(32) + (32)(32) + 32$

$P_{gate} = 1376$

For 4 gates : $P_{total} = 4P_{gate}$

$= 4(1376)$

$\{ P_{total} = 5504 \}$

(b) Two bias vectors per gate:

$P_{gate} = DH + H^2 + 2H$

$P_{gate} = 320 + 1024 + 64$
$= 1408$

For 4 gates : $P_{total} = 4(1408)$

$\{P_{total} = 5632\}$

(3) The gradient scaling is approximated by

Scaling factor $= f^k$

f : constant forgot gate value
k : no. of time steps

(a) $f = 0.9$

After 10 steps

$SF = 0.9^{10}$

$= 0.3486784401$

Rounded to 4 decimal places

0.3487 //

The gradient retains approx 34.87 % of its original magnitude

After 50 steps:

$SF = 0.9^{50}$

$= 0.0051538$

$= 0.0052$ //

The gradient retains approx 0.52-1. of its original magnitude

(b) $k = 50$

Case 1: $f = 0.5$

$SF = 0.5^{50}$

$= 8.8818 \times 10^{-16}$

$= 0.000$

The result is zero, although the actual value is extremely small & not exactly zero.

Case 2: $f = 0.99$

$SF = 0.99^{50}$

≈ 0.6050

The gradient retains approx 60.50-1. of its original magnitude

(c) Interpretation

When the forget gate is 0.5 on 0.9, the gradient becomes very small over 50 time steps, making it difficult for the network to learn long term dependencies. When the forget gate is 0.99 the gradient retains about 60.50-1. of its magnitude

(4) (a) Total hourly records

One day contains 24 hours
One year : $365 \times 24 = 8760$ hrs.

Three years : $N = 3 \times 8760$
$\{N = 26,280$ records$\}$

(b) no of input / output windows.

Each window requires :
168 input records
24 only next target records.

with a stride of 1 hours, the no. of valid window is :

window $= N -$ lookback $-$ horizon $+ 1$
$= 26280 - 168 - 24 + 1$
$= 26089$ //

The +1 accounts for the first valid window.

(c) Tensor shape for a batch of 64 windows.

The standard input shape is
(batch size, time steps, features)

$\{X = 64, 168, 3\}$

For 24 hrs forecast, assuming the model predicts one load value per hour.

$$\{y : 64, 24\}$$

This means:

- 64 separate windows
- each is containing 168 hourly records & 3 features per record.
- each op. contains 24 predicted hourly load values.

(d) Training window & mini-batches.

The first 70 -/. of records are used for training

Step 1: Calculate training records.

$$N_{train} = 0.70 \times 26280$$
$$= 18,396$$

Step 2 Calculate training windows

$$W_{train} = N_{train} - 168 - 24 + 1$$

$$= 18,205 //$$

Step-3 Calculate mini-batches

$$\text{Batches} = \frac{\text{Training window}}{\text{Batch Size}}$$

$$= \frac{18205}{64} \text{ //}$$

Since : $64 \times 284 = 18176$

Remaining window : $18205 - 18176 = 29$

There are 284 full batches & one final partial batch.

285 mini-batches per epoch.

(e) LSTM parameter count

Given input dim $D = 3$
Hidden size $H = 64$
One bias vector per gate

$P = 4H(D + H + 1)$
$P = 4(64)(3 + 64 + 1)$
$P = 17,408$ // trainable parameters

(5) (a) Min-Max Normalization

Given min reading = 20 mm/s
max " = 120 mm/s
actual " = 95 mm/s

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} = \frac{95 - 20}{120 - 20}$$

$$x_{norm} = 0.75 //$$

Normalised reading is 0.75, lies in range 0 & 1.

(6) failure prediction using threshold

Given:

Predicted failure prob = 0.83
Decision threshold = 0.5

Decision rule:

$$\hat{y} = \begin{cases} 1, & p \geq 0.5 \\ 0, & p < 0.5 \end{cases}$$

Compare

$0.83 \geq 0.5$

∴ Prediction = 1 (failure)

The model predicts that the pump will fail within the next cycle.

(c) Confusion matrix & evaluation matrix.

Given : total cycles = 1,000
actual failures = 40
predicted " = 45
correctly " = 30

TP = 30 //

FP = Pred. failure - TP
= 45 - 30
= 15 //

FN = Actual failure - TP
= 40 - 30
= 10 //

TN = 1000 - 40 - 15
= 945 //

Confusion Matrix

actual/predicted	Failure	No failure	total
failure	TP = 30	FN = 10	40
No failure	FP = 15	TN = 945	960
total	45	955	1000

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{30}{30+15} = 66.67\%$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{30}{30+10} = 75\%$$

$$\text{F-1 score} = \frac{2PR}{P+R} = \frac{2(30/45)(30/40)}{30/45 + 30/40}$$

$$= \frac{60}{85} = 0.7059\%$$

$$\text{Accuracy} = \frac{TP + TN}{N} = \frac{30 + 945}{1000} = 97.50\%$$

(b) The maintenance team should generally